

Machine Learning Midterm 1 Study Material

This document provides detailed, exam-style answers for the Machine Learning Midterm 1 important questions, covering Units I, II, and III. Each answer is designed to be comprehensive, ranging from 12-15 marks, and includes relevant diagrams for better understanding.

UNIT I: Machine Learning Fundamentals

1. Distinguish between Supervised and Unsupervised Learning. (12-15 Marks)

Introduction

Machine Learning (ML) is broadly categorized based on how the learning algorithm is trained. The two most prominent paradigms are **Supervised Learning** and **Unsupervised Learning**. The primary difference lies in the presence or absence of labeled data during the training process.

Supervised Learning

Supervised learning is a type of machine learning where the model is trained on a **labeled dataset**. This means that for every input data point, the corresponding output (target) is already known. The algorithm “learns” the mapping function from the input to the output.

- **Mechanism:** The model makes predictions on the training data and is corrected by the “teacher” (the labels). The process continues until the model achieves a desired level of accuracy.
- **Types:**
 - **Regression:** Predicting continuous values (e.g., house prices).

- **Classification:** Predicting discrete categories (e.g., Spam vs. Not Spam).
- **Example:** Predicting whether an email is spam based on historical emails labeled as “spam” or “ham”.

Unsupervised Learning

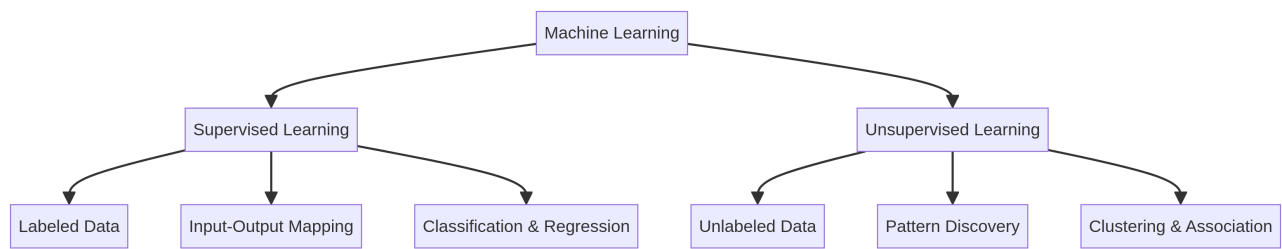
Unsupervised learning involves training a model on data that has **no labels**. The algorithm tries to find hidden patterns, structures, or relationships within the data without any external guidance.

- **Mechanism:** The model explores the data and identifies similarities or differences to group the data points.
- **Types:**
 - **Clustering:** Grouping similar data points together (e.g., customer segmentation).
 - **Association:** Finding rules that describe large portions of data (e.g., people who buy bread also buy butter).
 - **Dimensionality Reduction:** Reducing the number of variables while preserving important information.
- **Example:** A bank grouping its customers into different segments based on their spending habits without prior knowledge of the segments.

Key Differences Table

Feature	Supervised Learning	Unsupervised Learning
Data Type	Labeled Data	Unlabeled Data
Goal	Predict outcomes for new data	Discover hidden patterns/structures
Feedback	Direct feedback (labels)	No feedback
Complexity	Simpler (clear objective)	More complex (exploratory)
Algorithms	Linear Regression, SVM, Random Forest	K-Means, PCA, Apriori
Accuracy	High (measurable)	Lower (subjective evaluation)

Diagram: Supervised vs. Unsupervised Learning



2. Explain the Applications of Machine Learning. (12-15 Marks)

Introduction

Machine Learning has permeated almost every industry, transforming how businesses operate and how individuals interact with technology. Its ability to learn from data and make predictions makes it invaluable.

Key Applications

1. Healthcare and Medical Diagnosis

- **Disease Prediction:** ML models analyze patient history, lifestyle, and genetic data to predict the likelihood of diseases like diabetes or heart conditions.
- **Medical Imaging:** Algorithms assist radiologists in detecting tumors or fractures from X-rays, MRIs, and CT scans with high precision.
- **Drug Discovery:** ML accelerates the process of identifying potential drug candidates by simulating chemical interactions.

2. Finance and Banking

- **Fraud Detection:** Banks use ML to monitor transactions in real-time and flag suspicious activities that deviate from a user's normal behavior.
- **Algorithmic Trading:** ML models analyze market trends and execute trades at optimal times to maximize profit.
- **Credit Scoring:** Assessing the creditworthiness of loan applicants based on their financial history.

3. E-commerce and Marketing

- **Recommendation Systems:** Platforms like Amazon and Netflix use ML to suggest products or movies based on a user's past behavior.
- **Sentiment Analysis:** Analyzing social media posts and reviews to understand customer sentiment toward a brand.
- **Dynamic Pricing:** Adjusting prices in real-time based on demand, competition, and user profiles (e.g., Uber surge pricing).

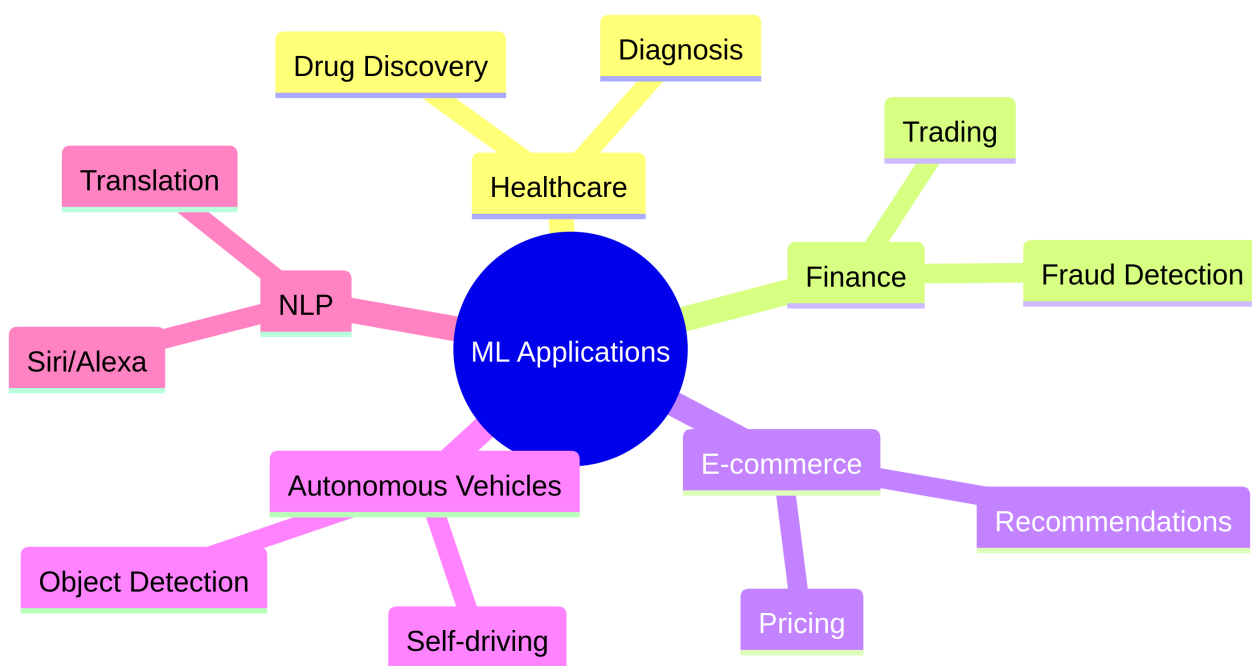
4. Autonomous Vehicles

- **Self-Driving Cars:** Companies like Tesla and Waymo use ML for object detection, lane tracking, and decision-making in real-time to navigate safely.

5. Natural Language Processing (NLP)

- **Virtual Assistants:** Siri, Alexa, and Google Assistant use ML to understand and respond to human speech.
- **Machine Translation:** Tools like Google Translate use deep learning to translate text between languages accurately.

Diagram: Applications of Machine Learning



3. Define Machine Learning. Demonstrate various stages in machine learning briefly. (12-15 Marks)

Definition of Machine Learning

Arthur Samuel (1959) defined Machine Learning as the “**field of study that gives computers the ability to learn without being explicitly programmed.**” Tom Mitchell (1997) provided a more formal definition: “**A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.**”

Stages in the Machine Learning Process

1. Data Collection

The first step is gathering relevant data from various sources like databases, sensors, or web scraping. The quality and quantity of data directly impact the model's performance.

2. Data Preprocessing

Raw data is often messy. This stage involves:

- **Data Cleaning:** Handling missing values and removing outliers.
- **Data Transformation:** Scaling, normalization, and encoding categorical variables.

3. Feature Engineering

Selecting the most relevant variables (features) that contribute to the prediction. This may involve creating new features from existing ones.

4. Model Selection

Choosing the appropriate algorithm (e.g., Decision Trees, Neural Networks) based on the problem type (classification, regression, etc.).

5. Training the Model

The algorithm is fed the training data to learn the underlying patterns. The model adjusts its internal parameters to minimize error.

6. Evaluation

The model's performance is tested on a separate **test dataset** using metrics like Accuracy, Precision, Recall, or Mean Squared Error.

7. Hyperparameter Tuning

Optimizing the model's settings (hyperparameters) to improve performance.

8. Deployment and Monitoring

The final model is integrated into a production environment and monitored for its performance on real-world data.

Diagram: Stages of Machine Learning



4. Classify various types of Machine Learning. Explain any one type of ML with diagrammatic representation. (12-15 Marks)

Classification of Machine Learning

Machine Learning is primarily classified into four types:

1. **Supervised Learning:** Learning with labeled data (e.g., Classification, Regression).
2. **Unsupervised Learning:** Learning without labels (e.g., Clustering, Association).
3. **Semi-Supervised Learning:** A mix of a small amount of labeled data and a large amount of unlabeled data.
4. **Reinforcement Learning:** Learning through trial and error based on rewards and penalties.

Detailed Explanation: Supervised Learning

Supervised learning is the most common type of ML. It works under the supervision of a “teacher” (the labels).

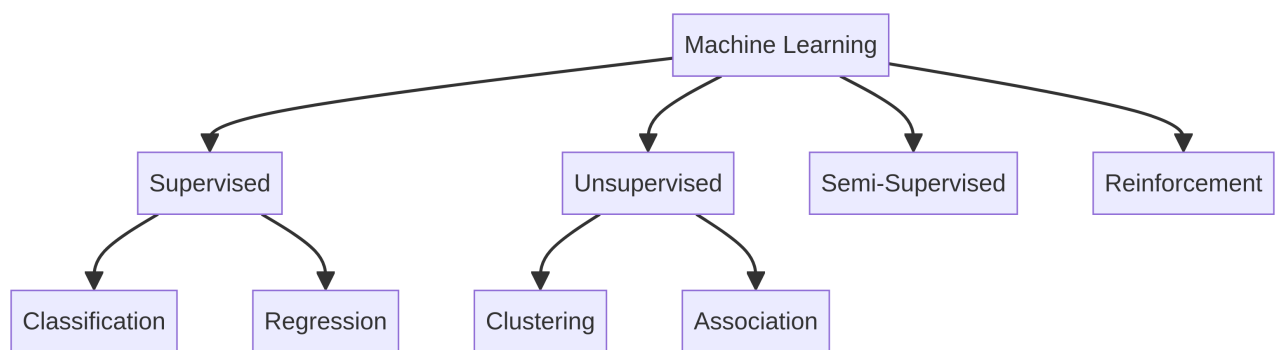
- **Process:**

1. Provide the algorithm with input-output pairs (x, y) .
2. The algorithm learns a function $f(x) = y$.
3. When new data x' is provided, the model predicts y' .

- **Sub-types:**

- **Classification:** The output is a category (e.g., “Yes” or “No”).
- **Regression:** The output is a continuous value (e.g., “75.5”).

Diagram: Classification of ML



5. Summarize the concepts of learning by Rote & learning by Induction with suitable examples. (12-15 Marks)

Learning by Rote (Memorization)

Learning by Rote is the simplest form of learning. It involves **storing information exactly as it is received** without any processing or generalization.

- **Mechanism:** The system memorizes the input-output pairs. When a known input is encountered, it simply retrieves the stored output.
- **Limitation:** It cannot handle new or unseen data. If the input is slightly different from what was memorized, the system fails.

- **Example:** A computer program that stores a multiplication table. It can tell you $5 \times 5 = 25$ because it's stored, but it cannot calculate 5.1×5.1 if it wasn't explicitly stored.

Learning by Induction (Generalization)

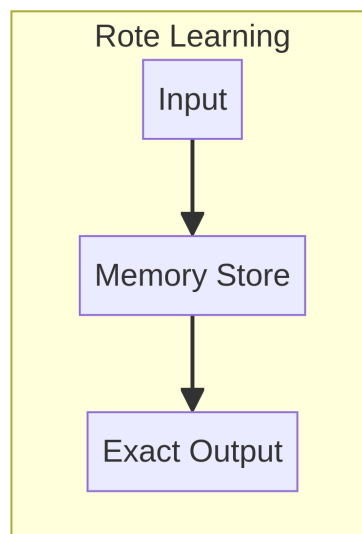
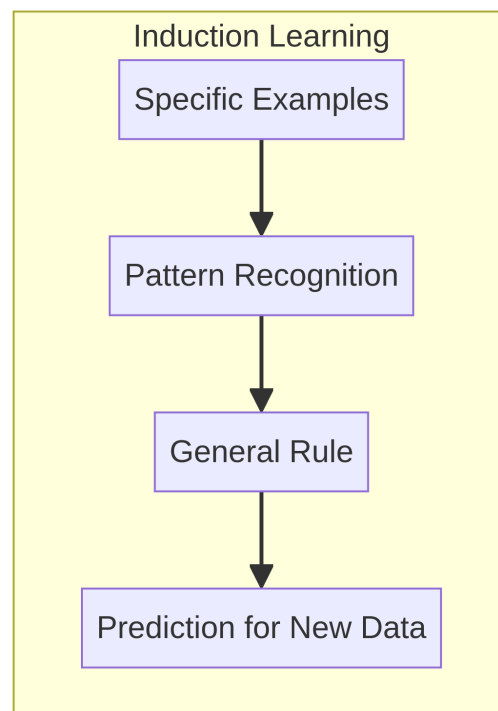
Learning by Induction is the process of **deriving general rules or patterns from specific examples**. This is the core of most modern Machine Learning.

- **Mechanism:** The system looks at many examples and finds a common pattern. It creates a model that can be applied to new, unseen data.
- **Strength:** It can generalize. It doesn't just memorize; it understands the underlying logic.
- **Example:** If a child sees several birds and notices they all have feathers and can fly, the child "induces" the general rule: "If an animal has feathers and flies, it is a bird."

Comparison Table

Feature	Learning by Rote	Learning by Induction
Core Concept	Memorization	Generalization
New Data	Fails on unseen data	Can predict for unseen data
Efficiency	Fast retrieval	Requires training time
Intelligence	Low (Static)	High (Adaptive)

Diagram: Rote vs. Induction



6. Demonstrate different ways to handle missing values in the dataset. (12-15 Marks)

Introduction

Missing data is a common problem in real-world datasets. It can occur due to sensor failure, human error, or data corruption. Handling missing values is crucial because

most ML algorithms cannot handle them directly.

Methods to Handle Missing Values

1. Deletion (Data Dropping)

- **Listwise Deletion:** Removing the entire row if any value is missing.
- **Column Deletion:** Removing the entire feature if it has too many missing values (e.g., > 50%).
- **Pros/Cons:** Simple but can lead to significant loss of information.

2. Imputation (Filling Values)

- **Mean/Median/Mode Imputation:** Replacing missing values with the average (mean), middle (median), or most frequent (mode) value of that column.
 - *Mean:* Good for numerical data without outliers.
 - *Median:* Better for numerical data with outliers.
 - *Mode:* Used for categorical data.
- **Constant Value Imputation:** Replacing missing values with a specific value like “Unknown” or “0”.

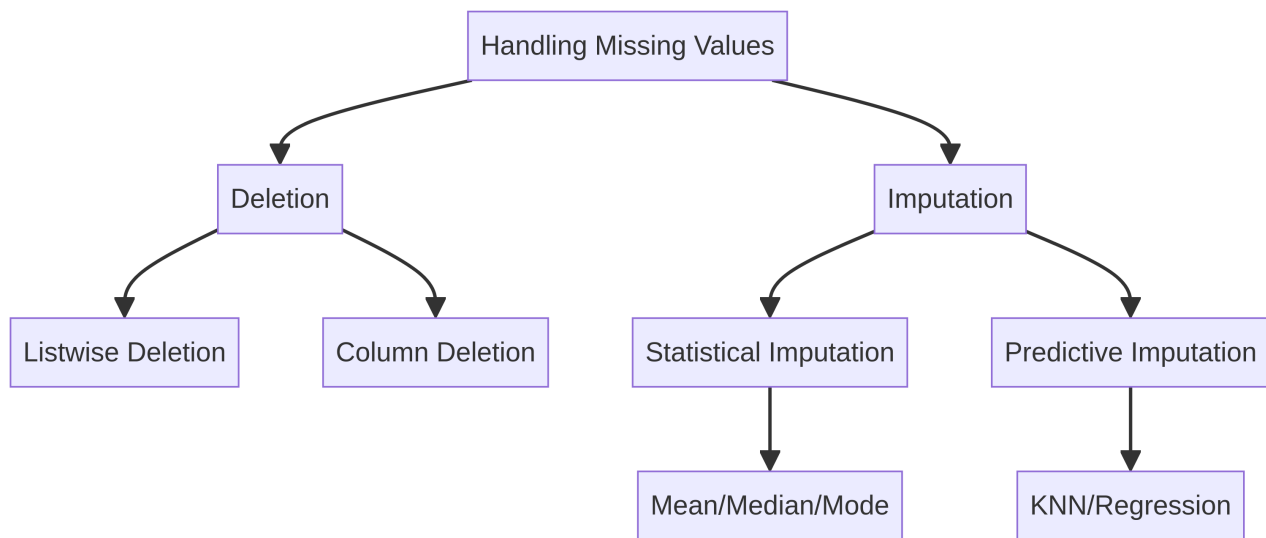
3. Predictive Imputation

- **K-Nearest Neighbors (KNN):** Finding the ‘K’ most similar rows and using their values to fill the gap.
- **Regression Imputation:** Using other features to predict the missing value via a regression model.

4. Using Algorithms that Support Missing Values

- Some algorithms like **XGBoost** or **LightGBM** can handle missing values internally during the splitting process.

Diagram: Handling Missing Values



7. Summarize Reinforcement Learning Algorithm. (12-15 Marks)

Introduction

Reinforcement Learning (RL) is a type of machine learning where an **Agent** learns to make decisions by performing actions in an **Environment** to maximize a **Cumulative Reward**.

Key Components of RL

1. **Agent:** The learner or decision-maker.
2. **Environment:** Everything the agent interacts with.
3. **Action (A):** What the agent does.
4. **State (S):** The current situation of the agent.
5. **Reward [®]:** Feedback from the environment (positive for good actions, negative for bad).
6. **Policy (π):** The strategy the agent uses to decide the next action based on the state.

How it Works (The RL Loop)

1. The agent observes the current **State**.

2. The agent selects an **Action** based on its **Policy**.
3. The environment transitions to a new state and gives a **Reward**.
4. The agent updates its knowledge to maximize future rewards.

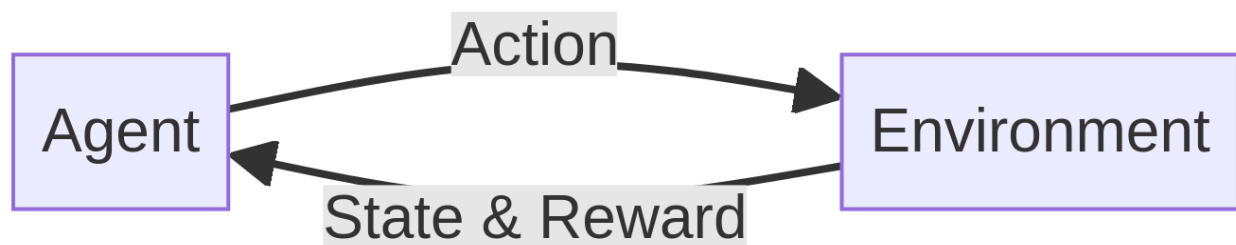
Exploration vs. Exploitation

- **Exploration:** Trying new actions to discover their rewards.
- **Exploitation:** Using known actions that yield high rewards.
- A good RL algorithm must balance these two.

Example

Training a robot to walk. If it takes a step forward without falling, it gets a +1 reward. If it falls, it gets a -10 reward. Over time, it learns the sequence of movements that keep it upright.

Diagram: Reinforcement Learning Loop



8. Examine Feature Subset Selection in Machine Learning with examples. (12-15 Marks)

Introduction

Feature Subset Selection (FSS) is the process of selecting a subset of relevant features for use in model construction. It helps in reducing overfitting, improving accuracy, and reducing training time.

Methods of Feature Selection

1. Filter Methods

These methods use statistical measures to score the correlation between each feature and the target variable.

- **Examples:** Chi-Square test, Information Gain, Correlation Coefficient.
- **Pros:** Fast and independent of the ML model.

2. Wrapper Methods

These methods use a specific ML model to evaluate subsets of features. They “wrap” the selection process around the model.

- **Forward Selection:** Start with no features and add one at a time.
- **Backward Elimination:** Start with all features and remove the least significant one at a time.
- **Pros:** More accurate but computationally expensive.

3. Embedded Methods

These methods perform feature selection during the model training process.

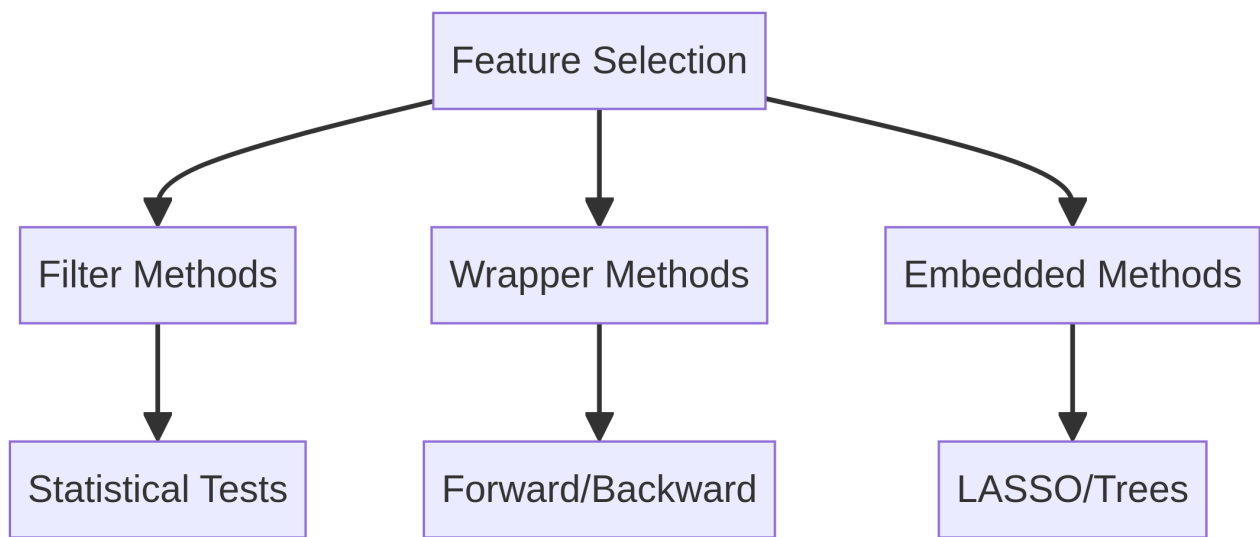
- **Examples:** LASSO Regression (L1 Regularization), Decision Trees (Feature Importance).
- **Pros:** Balance between filter and wrapper methods.

Example

In a dataset for predicting “House Price”, features might include “Square Footage”, “Number of Bedrooms”, “Color of the Front Door”, and “Owner’s Middle Name”.

- **Feature Selection** would keep “Square Footage” and “Number of Bedrooms” while discarding the door color and owner’s name as they are irrelevant.

Diagram: Feature Selection Methods



UNIT II: Proximity Measures and Performance Evaluation

1. Distance Calculation for $p=(4,2,5)$ and $q=(2,1,-7)$. (12-15 Marks)

Introduction

Distance metrics are used to measure the similarity or dissimilarity between two data points in a multi-dimensional space. Given two points $p = (p_1, p_2, p_3)$ and $q = (q_1, q_2, q_3)$, we calculate various distances.

Calculations

1. Euclidean Distance (L_2 Norm)

Formula: $d(p, q) = \sqrt{\sum (p_i - q_i)^2}$

- $d = \sqrt{(4 - 2)^2 + (2 - 1)^2 + (5 - (-7))^2}$
- $d = \sqrt{2^2 + 1^2 + 12^2} = \sqrt{4 + 1 + 144} = \sqrt{149} \approx 12.206$

2. City-block Distance (L_1 Norm / Manhattan Distance)

Formula: $d(p, q) = \sum |p_i - q_i|$

- $d = |4 - 2| + |2 - 1| + |5 - (-7)|$
- $d = 2 + 1 + 12 = 15$

3. Max Norm (L_∞ Norm / Chebyshev Distance)

Formula: $d(p, q) = \max(|p_i - q_i|)$

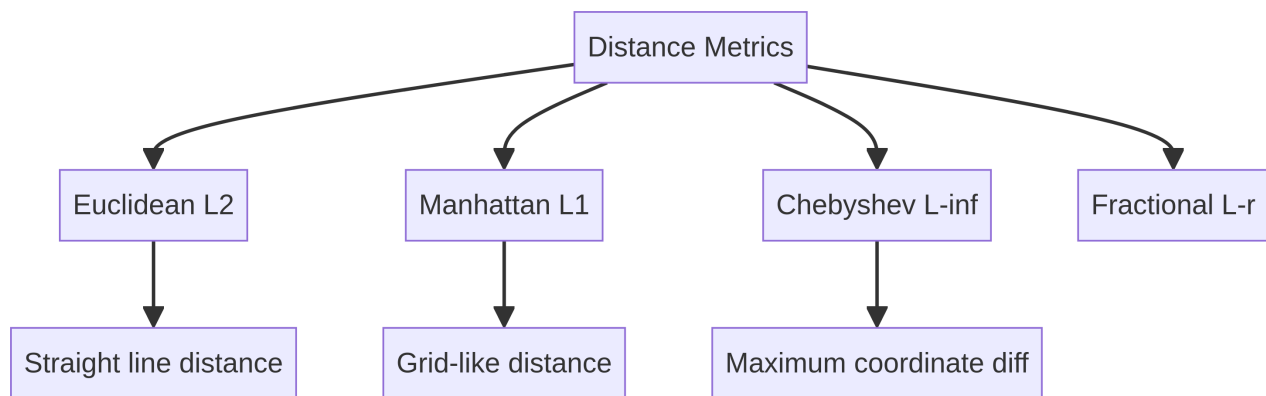
- $d = \max(|4 - 2|, |2 - 1|, |5 - (-7)|)$
- $d = \max(2, 1, 12) = 12$

4. Fractional Norm (L_r Norm where $r = 0.25$)

Formula: $d(p, q) = (\sum |p_i - q_i|^r)^{1/r}$

- $d = (|4 - 2|^{0.25} + |2 - 1|^{0.25} + |5 - (-7)|^{0.25})^{1/0.25}$
- $d = (2^{0.25} + 1^{0.25} + 12^{0.25})^4$
- $d = (1.189 + 1 + 1.861)^4 = (4.05)^4 \approx 269.06$

Diagram: Distance Metrics Visualization



2. Radius Based Nearest Neighbor Algorithm. (12-15 Marks)

Introduction

Radius-Based Nearest Neighbor (RNN) is a classification algorithm that classifies a test point based on the majority class of all training points within a fixed radius r .

Problem Data

- **Class 1:** (3.5,3), (4.5,1), (1,5), (2,6), (3.5,4), (4.5,3)
- **Class 2:** (2,3.5), (2.5,1), (3.5,2), (3,2), (2.5,2), (2,1)
- **Test Point:** (2.2, 5)

Step 1: Calculate Euclidean Distances to Test Point (2.2, 5)

1. (3.5, 3): $\sqrt{(3.5 - 2.2)^2 + (3 - 5)^2} = \sqrt{1.3^2 + (-2)^2} = \sqrt{1.69 + 4} = 2.38$
2. (4.5, 1): $\sqrt{(4.5 - 2.2)^2 + (1 - 5)^2} = \sqrt{2.3^2 + (-4)^2} = \sqrt{5.29 + 16} = 4.61$
3. (1, 5): $\sqrt{(1 - 2.2)^2 + (5 - 5)^2} = \sqrt{(-1.2)^2 + 0} = 1.2$
4. (2, 6): $\sqrt{(2 - 2.2)^2 + (6 - 5)^2} = \sqrt{(-0.2)^2 + 1^2} = \sqrt{0.04 + 1} = 1.02$
5. (3.5, 4): $\sqrt{(3.5 - 2.2)^2 + (4 - 5)^2} = \sqrt{1.3^2 + (-1)^2} = \sqrt{1.69 + 1} = 1.64$
6. (4.5, 3): $\sqrt{(4.5 - 2.2)^2 + (3 - 5)^2} = \sqrt{2.3^2 + (-2)^2} = \sqrt{5.29 + 4} = 3.05$
7. (2, 3.5): $\sqrt{(2 - 2.2)^2 + (3.5 - 5)^2} = \sqrt{(-0.2)^2 + (-1.5)^2} = \sqrt{0.04 + 2.25} = 1.51$
8. (2.5, 1): $\sqrt{(2.5 - 2.2)^2 + (1 - 5)^2} = \sqrt{0.3^2 + (-4)^2} = \sqrt{0.09 + 16} = 4.01$
9. (3.5, 2): $\sqrt{(3.5 - 2.2)^2 + (2 - 5)^2} = \sqrt{1.3^2 + (-3)^2} = \sqrt{1.69 + 9} = 3.27$
10. (3, 2): $\sqrt{(3 - 2.2)^2 + (2 - 5)^2} = \sqrt{0.8^2 + (-3)^2} = \sqrt{0.64 + 9} = 3.10$
11. (2.5, 2): $\sqrt{(2.5 - 2.2)^2 + (2 - 5)^2} = \sqrt{0.3^2 + (-3)^2} = \sqrt{0.09 + 9} = 3.01$
12. (2, 1): $\sqrt{(2 - 2.2)^2 + (1 - 5)^2} = \sqrt{(-0.2)^2 + (-4)^2} = \sqrt{0.04 + 16} = 4.00$

Case 1: Radius $r = 1.23$

Points within $r = 1.23$:

- (1, 5) - Distance 1.2 (Class 1)
- (2, 6) - Distance 1.02 (Class 1)

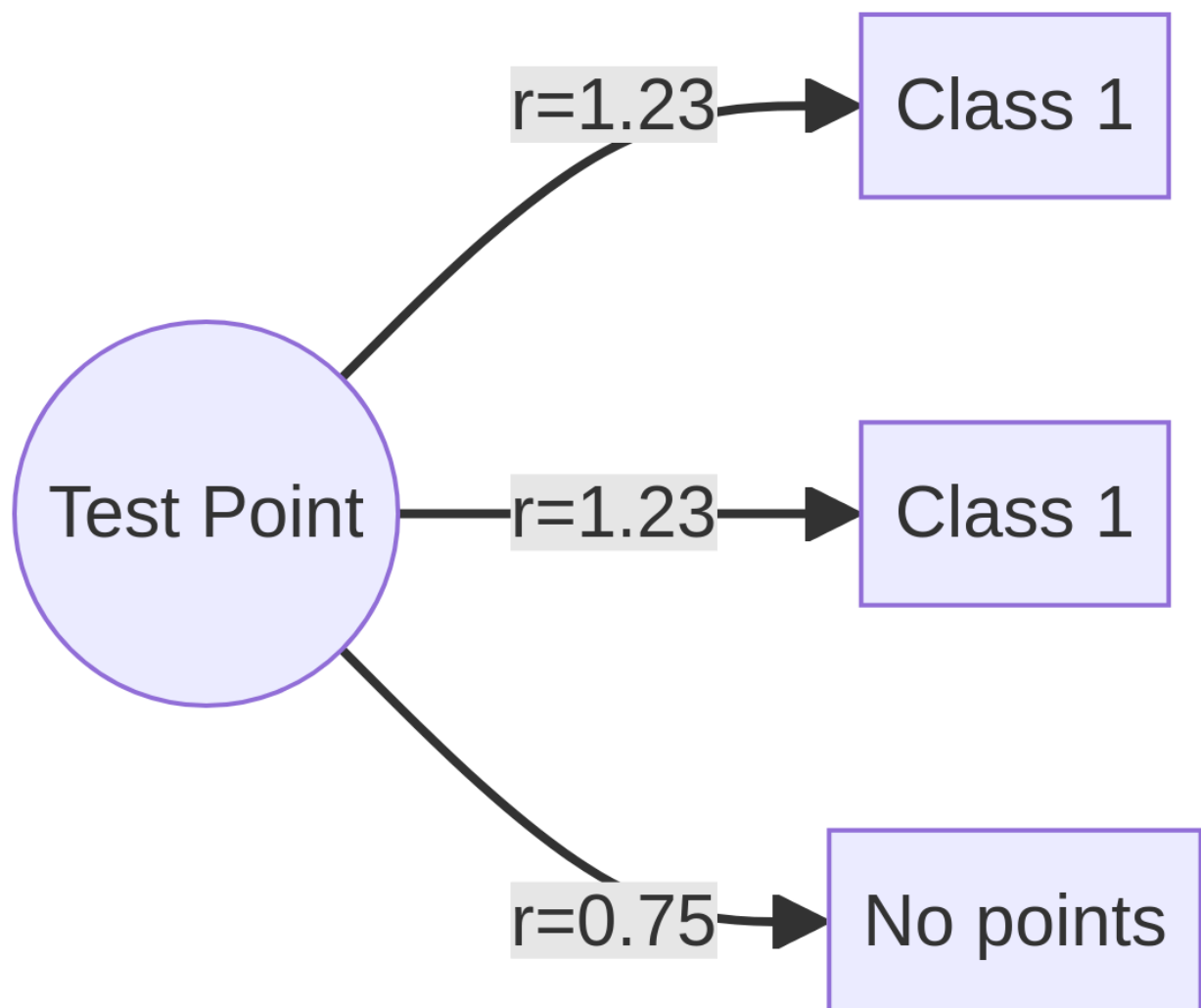
- **Result:** All points are Class 1. Test point (2.2, 5) is classified as **Class 1**.

Case 2: Radius $r = 0.75$

Points within $r = 0.75$:

- None.
- **Result:** The algorithm cannot classify the point (outlier or requires larger radius).

Diagram: Radius Based Classification



3. Proximity Measures on Binary Patterns. (12-15 Marks)

Introduction

Binary patterns are vectors containing only 0s and 1s. Proximity measures for binary data are based on the counts of matches and mismatches.

Given Patterns

- $p = 11001\ 00110\ 11011$
- $q = 00011\ 11100\ 01100$

Step 1: Count the Contingency Table

- n_{11} (Both 1): ($p=1, q=1$) at positions 5, 11. Count = 2.
- n_{10} ($p=1, q=0$): at positions 1, 2, 9, 13, 14, 15. Count = 6.
- n_{01} ($p=0, q=1$): at positions 4, 6, 7, 8, 12. Count = 5.
- n_{00} (Both 0): at positions 3, 10. Count = 2.
- Total $n = 15$.

Step 2: Calculate Measures

1. Simple Matching Coefficient (SMC)

Formula: $SMC = \frac{n_{11} + n_{00}}{n_{11} + n_{00} + n_{10} + n_{01}}$

- $SMC = \frac{2+2}{15} = \frac{4}{15} \approx 0.267$

2. Jaccard Coefficient (J)

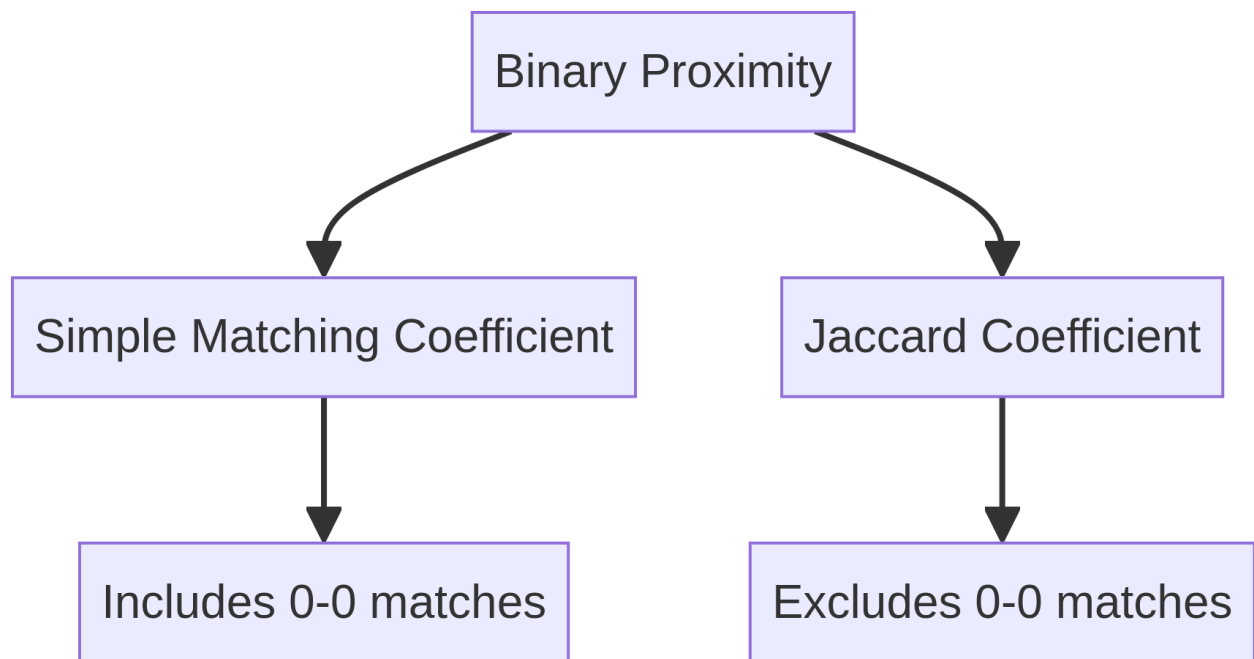
Formula: $J = \frac{n_{11}}{n_{11} + n_{10} + n_{01}}$

- $J = \frac{2}{2+6+5} = \frac{2}{13} \approx 0.154$

3. Tanimoto Coefficient

Often used interchangeably with Jaccard for binary data.

Diagram: Binary Proximity Table



4. KNN Algorithm for Bank Customer Classification. (12-15 Marks)

Introduction

K-Nearest Neighbors (KNN) classifies a point based on the majority class of its 'k' nearest neighbors.

Data

- **A:** (25, 30, 700, 10) - Low
- **B:** (45, 80, 600, 40) - High
- **C:** (35, 50, 750, 20) - Low
- **D:** (55, 90, 580, 50) - High
- **E:** (40, 60, 460, 25) - High
- **F (Test):** (32, 40, 500, 30)

Step 1: Calculate Manhattan Distance to F

$$d(F, X) = |Age_F - Age_X| + |Inc_F - Inc_X| + |CS_F - CS_X| + |Loan_F - Loan_X|$$

$$1. d(F, A) = |32 - 25| + |40 - 30| + |500 - 700| + |30 - 10| = 7 + 10 + 200 + 20 = 237$$

$$2. d(F, B) = |32 - 45| + |40 - 80| + |500 - 600| + |30 - 40| = 13 + 40 + 100 + 10 = 163$$

$$3. d(F, C) = |32 - 35| + |40 - 50| + |500 - 750| + |30 - 20| = 3 + 10 + 250 + 10 = 273$$

$$4. d(F, D) = |32 - 55| + |40 - 90| + |500 - 580| + |30 - 50| = 23 + 50 + 80 + 20 = 173$$

$$5. d(F, E) = |32 - 40| + |40 - 60| + |500 - 460| + |30 - 25| = 8 + 20 + 40 + 5 = 73$$

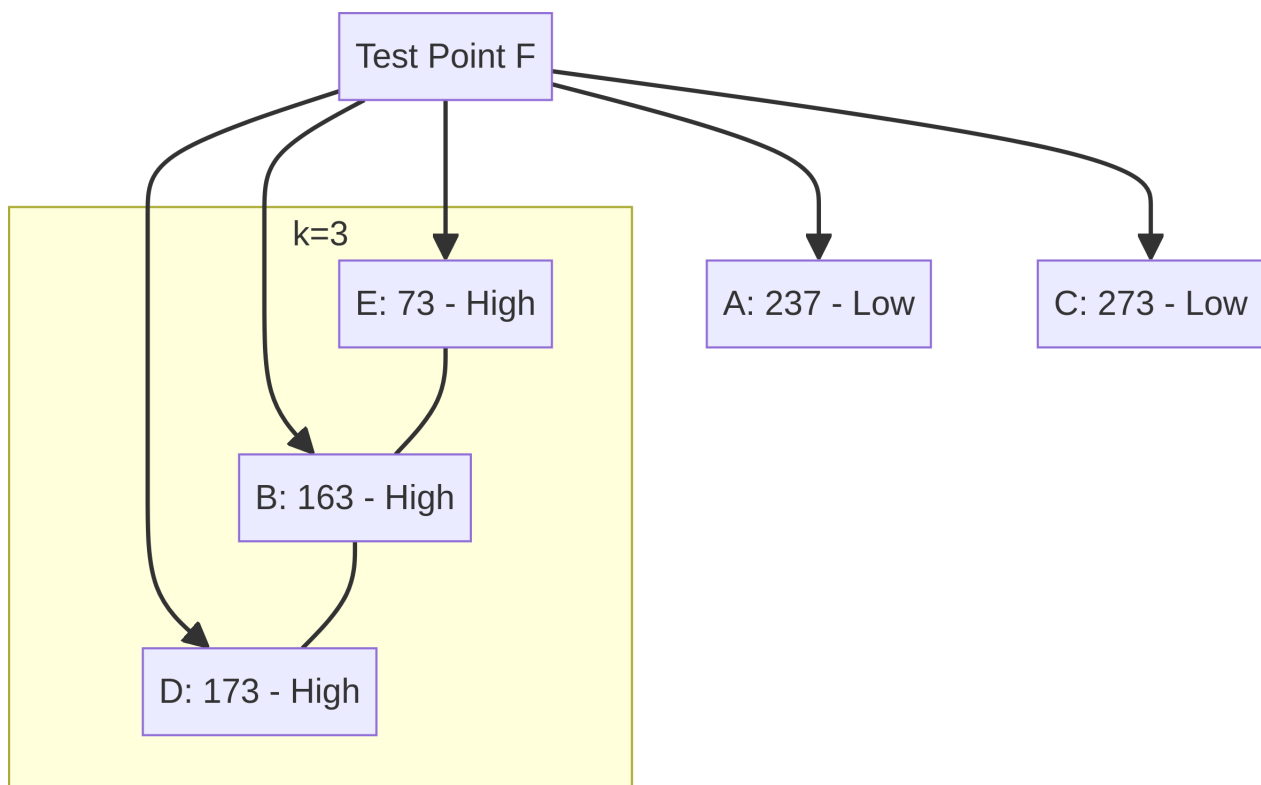
Step 2: Sort Distances

1. E: 73 (High)
2. B: 163 (High)
3. D: 173 (High)
4. A: 237 (Low)
5. C: 273 (Low)

Step 3: Classification

- **For k=3:** Neighbors are {E, B, D}. All are “High”. Result: **High-Risk**.
- **For k=4:** Neighbors are {E, B, D, A}. 3 High, 1 Low. Result: **High-Risk**.

Diagram: KNN Classification



5. Non-Metric Proximity Measures. (12-15 Marks)

Introduction

A distance metric must satisfy four properties: Non-negativity, Identity, Symmetry, and Triangle Inequality. A **Non-Metric Proximity Measure** is a measure that violates at least one of these properties (usually the Triangle Inequality).

Properties of a Metric

1. $d(x, y) \geq 0$
2. $d(x, y) = 0$ iff $x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle Inequality)

Example: Cosine Similarity

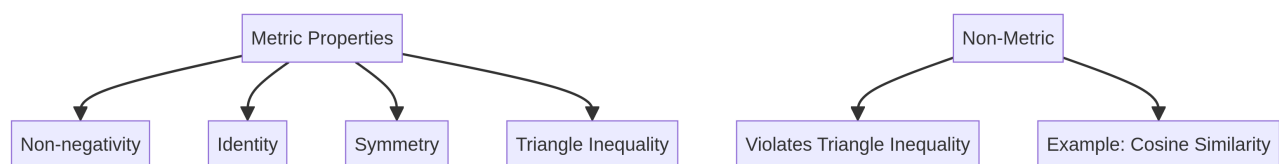
Cosine similarity measures the cosine of the angle between two vectors. It is widely used in text mining.

- **Formula:** $\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$
- **Why Non-Metric?** It does not satisfy the triangle inequality. Also, the “distance” version $(1 - \cos(\theta))$ is often used, but it still violates metric properties.

Example: Squared Euclidean Distance

While Euclidean distance is a metric, its square is not because it violates the triangle inequality.

Diagram: Metric vs Non-Metric



6. Classification Performance Evaluation Measures. (12-15 Marks)

Introduction

Evaluating a classification model is essential to understand its effectiveness. Accuracy alone is often misleading, especially in imbalanced datasets.

Key Measures

1. Confusion Matrix

A table used to describe the performance of a classification model.

- **TP (True Positive):** Correctly predicted positive.
- **TN (True Negative):** Correctly predicted negative.
- **FP (False Positive):** Incorrectly predicted positive (Type I Error).

- **FN (False Negative):** Incorrectly predicted negative (Type II Error).

2. Accuracy

The ratio of correctly predicted observations to the total observations.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$

3. Precision (Specificity)

The ratio of correctly predicted positive observations to the total predicted positives.

- $Precision = \frac{TP}{TP+FP}$

4. Recall (Sensitivity)

The ratio of correctly predicted positive observations to all observations in the actual class.

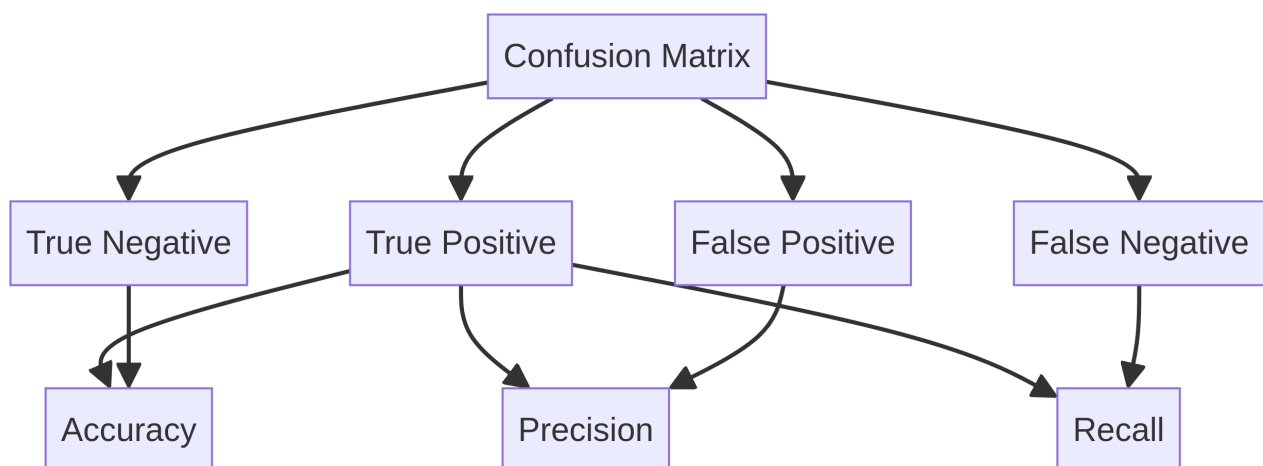
- $Recall = \frac{TP}{TP+FN}$

5. F1-Score

The weighted average of Precision and Recall.

- $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$

Diagram: Confusion Matrix and Metrics



7. Confusion Matrix Analysis. (12-15 Marks)

Given Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	100 (TP)	25 (FN)
Actual No	18 (FP)	65 (TN)

Performance Measures

1. Accuracy

- $Acc = \frac{100+65}{100+65+18+25} = \frac{165}{208} \approx 0.793 \text{ (79.3\%)}$

2. Precision

- $Prec = \frac{100}{100+18} = \frac{100}{118} \approx 0.847 \text{ (84.7\%)}$

3. Recall

- $Rec = \frac{100}{100+25} = \frac{100}{125} = 0.80 \text{ (80\%)}$

4. F1-Score

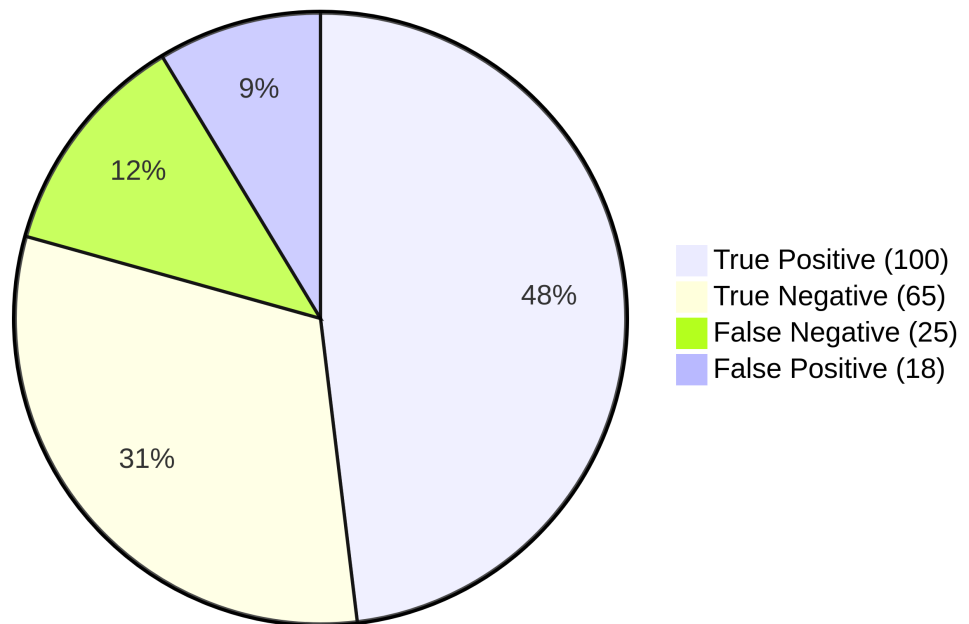
- $F1 = 2 \times \frac{0.847 \times 0.80}{0.847 + 0.80} = 2 \times \frac{0.6776}{1.647} \approx 0.823$

Conclusion

The model has a high precision (84.7%), meaning when it predicts “Yes”, it is likely correct. The overall accuracy is 79.3%.

Diagram: Visualizing the Matrix

Confusion Matrix Distribution



8. Regression Performance Evaluation Measures. (12-15 Marks)

Introduction

Regression models predict continuous values. Evaluation measures focus on the difference between predicted and actual values (residuals).

Key Measures

1. **Mean Absolute Error (MAE):** Average of absolute differences.
2. **Mean Squared Error (MSE):** Average of squared differences.
3. **Root Mean Squared Error (RMSE):** Square root of MSE.
4. **R-Squared (R^2):** Proportion of variance explained by the model.

Test on Given Data

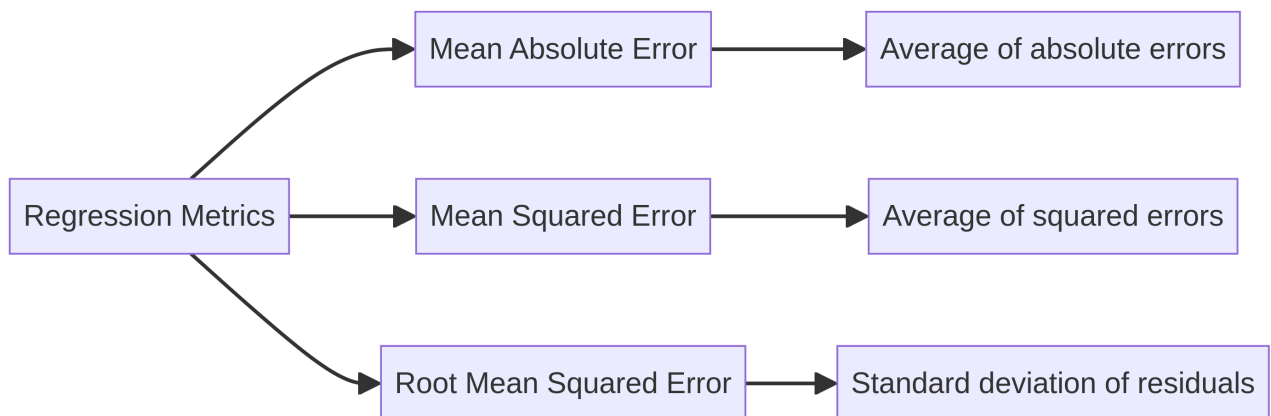
Hours	Actual (Y)	Predicted ($\hat{Y}$)	Error ($Y - \hat{Y}$)	$ Error $	$Error^2$
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2	60	63	-3	3	9
4	75	72	3	3	9
5	80	78	2	2	4
7	90	92	-2	2	4
6	89	90	-1	1	1
3	63	55	8	8	64
Total				19	91

Calculations

- **MAE** = $19/6 \approx 3.167$
- **MSE** = $91/6 \approx 15.167$
- **RMSE** = $\sqrt{15.167} \approx 3.894$

Diagram: Regression Error Visualization



UNIT III: Decision Tree Classification

1. Decision Tree Root Node Selection using Weighted Entropy. (12-15 Marks)

Introduction

The root node of a decision tree is selected based on the feature that provides the highest **Information Gain** (or the lowest **Weighted Entropy**).

Dataset Analysis

Total Samples: 14

- **Play? Yes:** 9
- **Play? No:** 5

Step 1: Calculate Total Entropy $H(S)$

$$H(S) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\ -1.485 = 0.410 + 0.530 = 0.940$$

$$H(S) = -(0.643 \times -0.637) - (0.357 \times$$

Step 2: Calculate Weighted Entropy for each Feature

1. Outlook

- **Sunny:** 5 samples (2 Yes, 3 No) $\rightarrow H(\text{Sunny}) = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) = 0.971$
- **Overcast:** 4 samples (4 Yes, 0 No) $\rightarrow H(\text{Overcast}) = 0$
- **Rainy:** 5 samples (3 Yes, 2 No) $\rightarrow H(\text{Rainy}) = 0.971$
- **Weighted Entropy (Outlook)** $= \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.347 + 0 + 0.347 = 0.694$

2. Temperature

- **Hot:** 4 samples (2 Yes, 2 No) $\rightarrow H(\text{Hot}) = 1.0$
- **Mild:** 6 samples (4 Yes, 2 No) $\rightarrow H(\text{Mild}) = 0.918$
- **Cool:** 4 samples (3 Yes, 1 No) $\rightarrow H(\text{Cool}) = 0.811$

- **Weighted Entropy (Temp)** = $\frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.286 + 0.393 + 0.232 = 0.911$

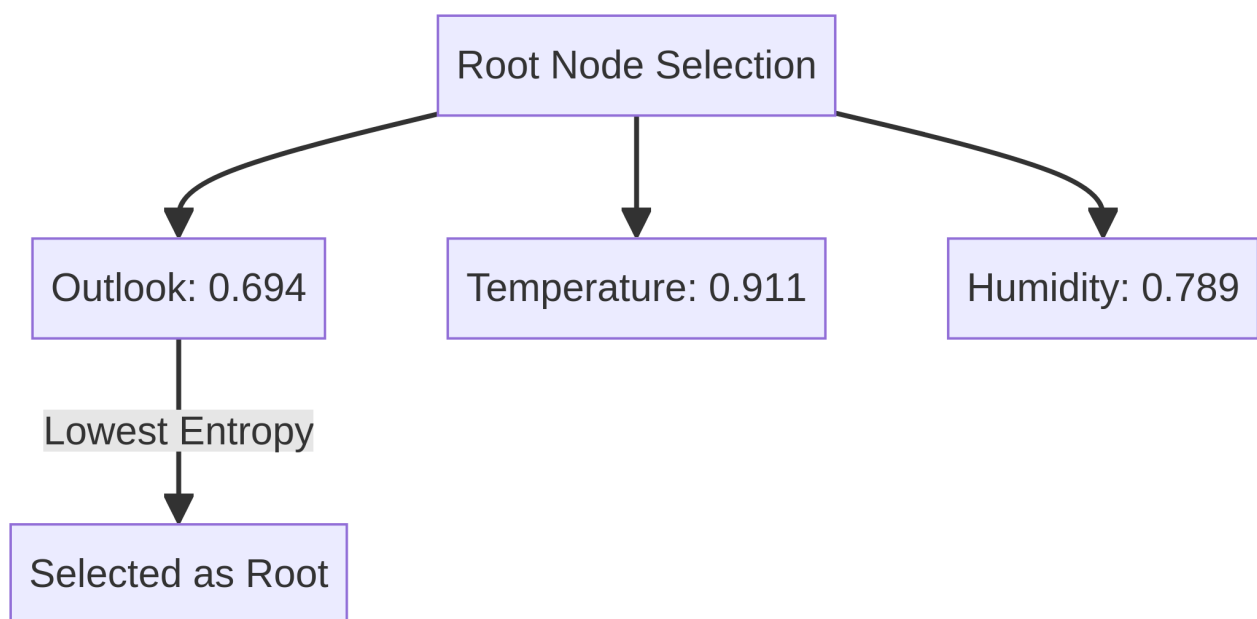
3. Humidity

- **High:** 7 samples (3 Yes, 4 No) $\rightarrow H(High) = 0.985$
- **Normal:** 7 samples (6 Yes, 1 No) $\rightarrow H(Normal) = 0.592$
- **Weighted Entropy (Humidity)** = $\frac{7}{14}(0.985) + \frac{7}{14}(0.592) = 0.493 + 0.296 = 0.789$

Conclusion

The feature with the **lowest weighted entropy** is **Outlook** (0.694). Therefore, **Outlook** is selected as the root node.

Diagram: Decision Tree Root Selection



2. Define Impurity and Analyze Different Measures. (12-15 Marks)

Definition of Impurity

In the context of Decision Trees, **Impurity** is a measure of how “mixed” the classes are in a given node. A node is “pure” if all its samples belong to the same class. A node is “impure” if it contains a mix of different classes.

Measures of Impurity

1. Entropy

Entropy measures the amount of uncertainty or randomness in the data.

- **Formula:** $H(S) = - \sum p_i \log_2(p_i)$
- **Range:** 0 (pure) to 1 (max impurity for binary).

2. Gini Index

Gini Index measures the probability of a random sample being incorrectly classified if it were labeled according to the distribution of labels in the subset.

- **Formula:** $Gini = 1 - \sum p_i^2$
- **Range:** 0 (pure) to 0.5 (max impurity for binary).

3. Classification Error

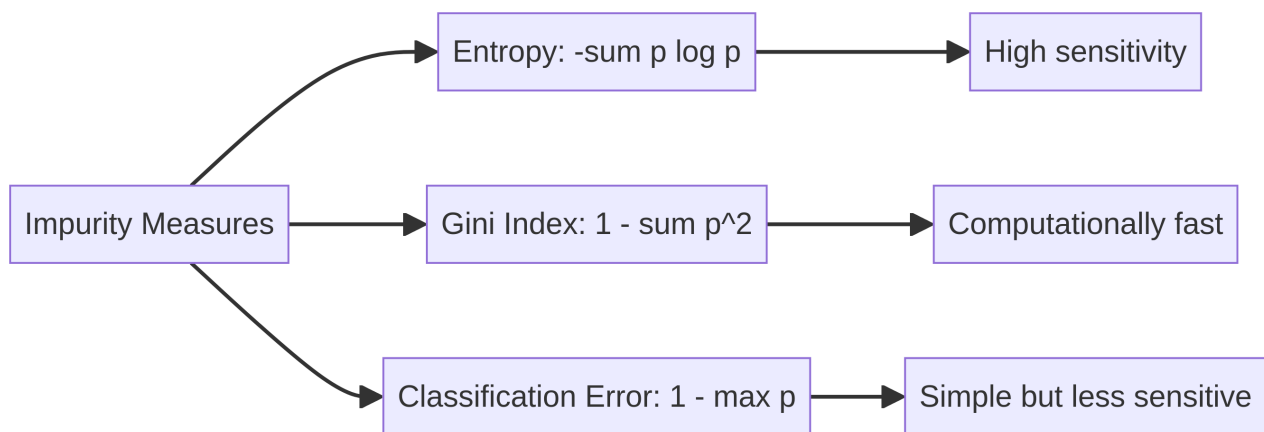
The simplest measure of impurity.

- **Formula:** $Error = 1 - \max(p_i)$

Comparison

- **Gini Index** is computationally faster as it doesn't involve logarithms.
- **Entropy** is more sensitive to changes in class probabilities.
- Both usually lead to similar trees.

Diagram: Impurity Measures Comparison



3. Decision Tree Root Node Selection using Gini Index. (12-15 Marks)

Introduction

The Gini Index is another popular measure for splitting nodes in a decision tree (used by the CART algorithm).

Dataset Analysis

Total Samples: 14 (9 Yes, 5 No)

- **Total Gini** = $1 - \left[\left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2\right] = 1 - [0.413 + 0.128] = 0.459$

Step 1: Calculate Weighted Gini for each Feature

1. Outlook

- **Sunny:** (2Y, 3N) $\rightarrow Gini = 1 - \left[\left(\frac{2}{5}\right)^2 + \left(\frac{3}{5}\right)^2\right] = 1 - [0.16 + 0.36] = 0.48$
- **Overcast:** (4Y, 0N) $\rightarrow Gini = 1 - [1^2 + 0^2] = 0$
- **Rainy:** (3Y, 2N) $\rightarrow Gini = 1 - \left[\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2\right] = 0.48$
- **Weighted Gini (Outlook)** = $\frac{5}{14}(0.48) + \frac{4}{14}(0) + \frac{5}{14}(0.48) = 0.171 + 0 + 0.171 = 0.342$

2. Temperature

- **Hot:** (2Y, 2N) $\rightarrow Gini = 0.5$

- **Mild:** (4Y, 2N) $\rightarrow Gini = 1 - [(\frac{4}{6})^2 + (\frac{2}{6})^2] = 0.444$
- **Cool:** (3Y, 1N) $\rightarrow Gini = 1 - [(\frac{3}{4})^2 + (\frac{1}{4})^2] = 0.375$
- **Weighted Gini (Temp)** $= \frac{4}{14}(0.5) + \frac{6}{14}(0.444) + \frac{4}{14}(0.375) = 0.143 + 0.190 + 0.107 = 0.440$

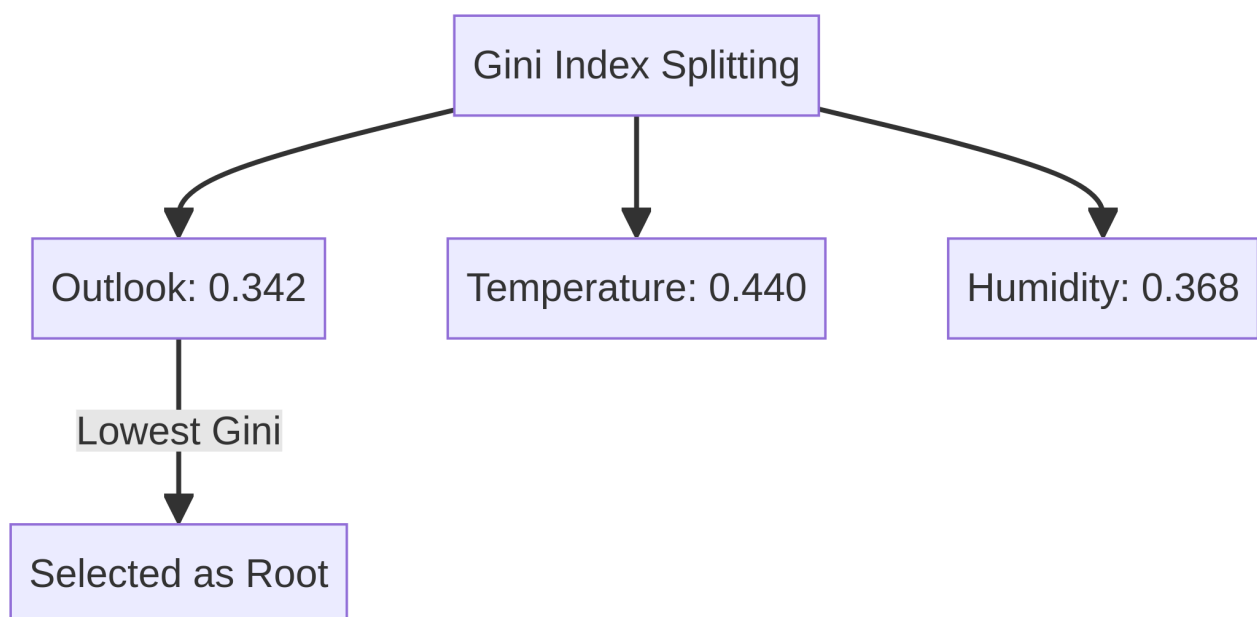
3. Humidity

- **High:** (3Y, 4N) $\rightarrow Gini = 1 - [(\frac{3}{7})^2 + (\frac{4}{7})^2] = 0.490$
- **Normal:** (6Y, 1N) $\rightarrow Gini = 1 - [(\frac{6}{7})^2 + (\frac{1}{7})^2] = 0.245$
- **Weighted Gini (Humidity)** $= \frac{7}{14}(0.490) + \frac{7}{14}(0.245) = 0.245 + 0.123 = 0.368$

Conclusion

The feature with the **lowest weighted Gini** is **Outlook** (0.342). Therefore, **Outlook** is selected as the root node.

Diagram: Gini Index Splitting



4. Outline the Properties of Decision Tree Classifier. (12-15 Marks)

Introduction

A Decision Tree is a non-parametric supervised learning method used for classification and regression. It has a flowchart-like structure where each internal node represents a “test” on an attribute.

Key Properties

1. Interpretability

Decision trees are highly intuitive and easy to explain. They can be visualized as a set of “if-then” rules.

2. Non-Parametric

They do not make any assumptions about the underlying distribution of the data (unlike Linear Regression which assumes normality).

3. Handles Both Data Types

Decision trees can handle both numerical (continuous) and categorical (discrete) data without much preprocessing.

4. Feature Selection

The algorithm naturally performs feature selection by placing the most important features at the top of the tree.

5. Sensitivity to Noise

Decision trees are prone to **overfitting**, especially if the tree is allowed to grow very deep. Small changes in the data can result in a completely different tree.

6. Computational Efficiency

Once built, the prediction process is very fast as it only involves a few comparisons.

7. Robustness to Outliers

Since they use binning/splitting rather than distance-based calculations, they are relatively robust to outliers.

Diagram: Decision Tree Structure

